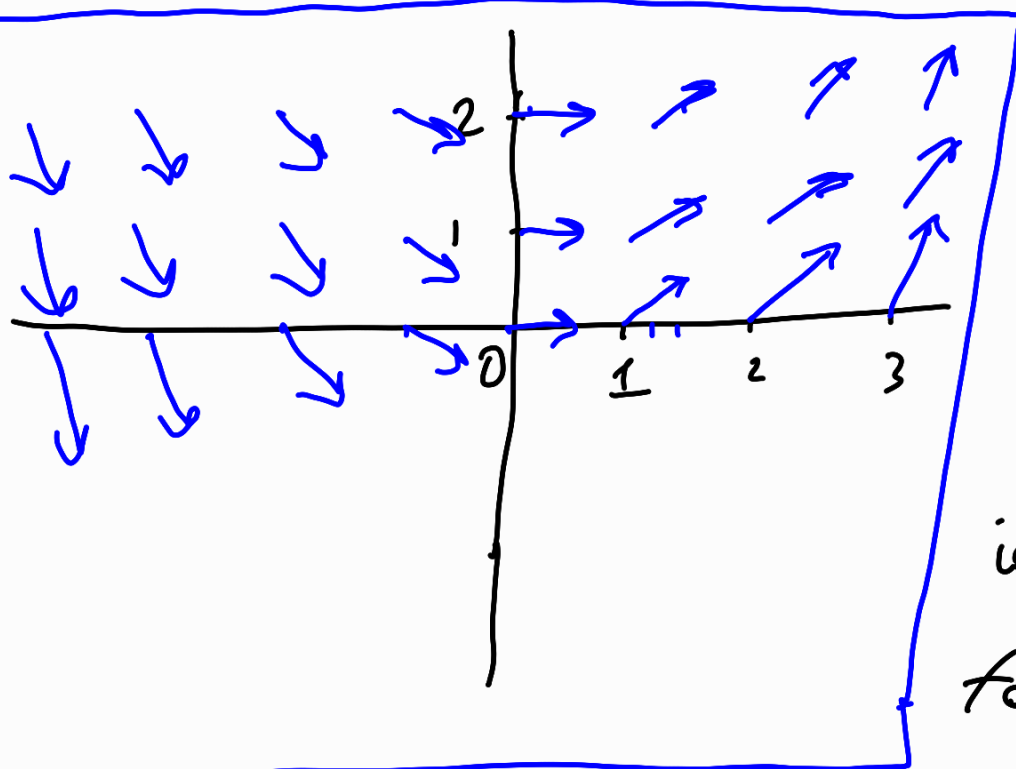


DW10 #8: Simpler way + note on difficulty provided in this note

No other announcements this week

1: Sketch $F(x, y) = \langle 2, x \rangle$

At any point (x, y) on the plane, draw the vector $\langle 2, x \rangle$. We can't



draw all
 ∞ -many
vectors, so
pick the points
in $\mathbb{Z}^2$. But
for the sake

of readability, all drawn vectors are
scaled down by $\approx \frac{1}{4}$ to not
overlap with each other.

2a: Find ∇f for $f = 2y \sin(xy)$

This is same ∇ from Lagrange mult.

$$f_x = 2y[\sin(xy)]_x = 2y^2 \cos(xy)$$

$$f_y = 2\sin(xy) + 2y[\sin(xy)]_y = 2\sin(\dots) + 2xy\cos(\dots)$$

$$\text{So } \nabla f = \langle f_x, f_y \rangle = \langle 2y^2 \cos(xy), 2\sin(xy) + 2xy \cos(xy) \rangle$$

4b: Find $\int_C f ds$, $f = x + y$, C is semicircle $x^2 + y^2 = 4$ with $y \geq 0$

Step 1: Parametrize C & find ds

Recall $r(t) = (2\cos t, 2\sin t)$, $0 \leq t \leq 2\pi$ would work for the full circle.

As t goes from 0 to π , the upper half of the circle is traced.

So we need $0 \leq t \leq \pi$.

Recall: $\frac{ds}{dt} = |r'(t)| = |2(-\sin t, \cos t)| =$
 $2\sqrt{\sin^2 t + \cos^2 t} = 2 \Rightarrow$ $ds = 2dt$

Step 2: Substitute into integral & evaluate

$$f = x+y = 2(\cos t + \sin t), \quad ds = 2dt,$$

$$\text{so } \int_C f ds = 4 \int_0^{\pi} (\cos t + \sin t) dt =$$

$$4(\sin t - \cos t) \Big|_0^{\pi} = 4 \cdot 2 = 8.$$

3a: Find if $F = \nabla f$ for some f for
 $F = \langle 4x^3y^3, 3x^4y^2 - 4y^3 \rangle$.

If $F = \nabla f = \langle f_x, f_y \rangle$ for some f ,
 notice that $(F_1)_y = (f_x)_y = (f_y)_x = (F_2)_x$

$$(F_1)_y = 12x^3y^2$$

$$(F_2)_x = (3x^4y^2 - 4y^3)_x = 12x^3y^2, \text{ so}$$

the existence of f has not been
 ruled out $\Rightarrow$ we can probably expect
 such an f to exist.

Note: If $F: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a continuously
 differentiable vector field and $\frac{\partial F_i}{\partial e_j} = \frac{\partial F_j}{\partial e_i}$
 for all coordinate choices (i, j) , then

there exists f such that $df = F$

$$f = \int f_x dx = \int 4x^3 y^3 dx = x^4 y^3 + g(y)$$

$$\text{So } 3x^4 y^2 + g'(y) = f_y = 3x^4 y^2 - 4y^3$$

$$\Rightarrow g'(y) = -4y^3 \Rightarrow g(y) = -y^4 + C.$$

The problem just says find $\textcolor{red}{a}$ function,
so we can choose $C=0$. Then

$$f = x^4 y^3 + g(y) = \textcolor{green}{x^4 y^3 - y^4} \text{ works.}$$

DW10 #8: Volume betw. $z = x^2 + y^2$ & $x^2 + y^2 + z^2 = 2$

It turns out cylindrical is cleaner than
spherical. Graphing the region & finding

$$\textcolor{green}{x^2 + y^2 + z^2 \leq 2} \text{ \& \& } \textcolor{green}{z \geq x^2 + y^2}$$

is still done the same way as before

Step 1: Use cylindrical coordinates

$$x = r \cos \theta, y = r \sin \theta, z = z \Rightarrow r^2 + z^2 \leq 2$$

and $z \geq r^2$. Notice $r^2 + z^2 \leq 2 \Rightarrow z \leq \sqrt{2 - r^2}$,

so $\textcolor{green}{r^2 \leq z \leq \sqrt{2 - r^2}}$. Naturally $\textcolor{green}{r \geq 0}$, but

$$\text{also } r^2 \leq \sqrt{2 - r^2} \Rightarrow f(r) := r^2 + r^4 \leq 2$$

$\Rightarrow$ $r \leq 1$ since f increasing & $f(1) = 2$.

Region rotationally symmetric, so $0 \leq \theta \leq 2\pi$

Step 2: Substitute bounds + dV & evaluate

Recall $dV = r dr d\theta dz$ for cyl. Then

$$\begin{aligned} V &= \iiint_V dV = \int_0^1 \int_0^{2\pi} \int_{r^2}^{\sqrt{2-r^2}} r dz dr d\theta = \\ &= 2\pi \int_0^1 r (\sqrt{2-r^2} - r^2) dr = 2\pi \int_0^1 r (2-r^2)^{1/2} - r^3 dr = \\ &= 2\pi \left[-\frac{1}{3} (2-r^2)^{3/2} - \frac{1}{4} r^4 \right] \Big|_0^1 = \\ &= 2\pi \left(-\frac{1}{3} - \frac{1}{4} + \frac{1}{3} 2^{3/2} \right) = \frac{2\pi}{3} \left(2\sqrt{2} - \frac{7}{4} \right) = \end{aligned}$$

$\frac{\pi}{6} (8\sqrt{2} - 7)$, same as with spherical
(from different instructor)

Note: Integrals at University of Delaware
math 243 will never require piecewise
bounds. Check your algebra, and if that's good,
try a different substitution. Piecewise needed
only in other universities/places.

4a: Find $\int_C f ds$, $f = \frac{x}{y}$, C given by
 $x = t^3$, $y = t^4$, $1 \leq t \leq 2$.

Step 1: Recognize $r(t)$ & find ds .

$$\underline{r(t) = \langle t^3, t^4 \rangle, \quad 1 \leq t \leq 2.}$$

Recall $\frac{ds}{dt} = |r'(t)| = |\langle 3t^2, 4t^3 \rangle| =$
 $|t^2 \langle 3, 4t \rangle| = t^2 |\langle 3, 4t \rangle| = t^2 \sqrt{9 + 16t^2},$

so $\underline{ds = t^2 \sqrt{9 + 16t^2} dt}$

Step 2: Substitute for f, ds , bounds

$$f = \frac{x}{y} = \frac{t^3}{t^4} = \frac{1}{t}, \text{ so}$$

$$f ds = t \sqrt{9 + 16t^2} dt, \text{ so}$$

$$\int_C f ds = \int_1^2 t (9 + 16t^2)^{1/2} dt$$

Step 3: Evaluate the integral

$$u = 9 + 16t^2 \Rightarrow 32t dt = du, \text{ and}$$

$$1 \leq t \leq 2 \Rightarrow \underset{25}{73} \leq u \leq 73, \text{ so}$$

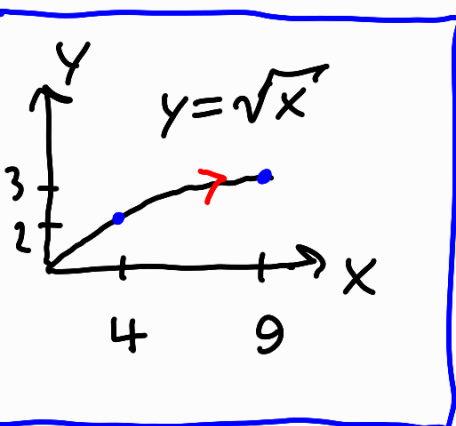
$$\text{int} = \int_{25}^{73} \frac{1}{32} u^{1/2} du = \frac{1}{32} \frac{2}{3} u^{3/2} \Big|_{25}^{73} =$$

$$\underline{\frac{1}{48} (73\sqrt{73} - 125)}.$$

2b: $f = x^3 y e^{y/2}$, $\nabla f =$
 $\langle 3x^2 y e^{y/2}, x^3(1 + \frac{y}{2})e^{y/2}, x^3 y(-\frac{y}{2^2})e^{y/2} \rangle$
 $= -\frac{x^3 y^2}{2^2} e^{y/2}$

4c: Find $\int_C f ds$, $f = xy$, C is arc of $y = \sqrt{x}$ from $(4, 2)$ to $(9, 3)$

Step 1: Parametrize C to find ds .



To begin, set $x = t$. Then from the graph, $4 \leq t \leq 9$.
 Also, $y = \sqrt{x} = \sqrt{t}$. So
 $r(t) = \langle t, \sqrt{t} \rangle$, $4 \leq t \leq 9$.

Recall: $\frac{ds}{dt} = |r'(t)| = |\langle 1, \frac{1}{2\sqrt{t}} \rangle| =$
 $\sqrt{1 + \frac{1}{4t}} = \frac{1}{2t} \sqrt{4t^2 + t}$.

Step 2: Substitute for r, ds, f
 $f = xy = t\sqrt{t}$, so $f ds = \frac{\sqrt{t}}{2} \sqrt{4t^2 + t} dt$
 $= \frac{1}{2} \sqrt{4t^3 + t^2} = \frac{t}{2} \sqrt{4t + 1} dt$, so

$$\int_C f ds = \int_4^9 \frac{1}{2} t (4t+1)^{1/2} dt$$

$$u = 4t+1 \Rightarrow du = 4dt, \quad 4 \leq t \leq 9 \Rightarrow 17 \leq u \leq 37,$$

$$\text{and } t = \frac{u-1}{4}, \text{ so } \frac{1}{2} t dt = \frac{1}{2} \frac{u-1}{4} \frac{du}{4}.$$

$$\text{int} = \int_{17}^{37} \frac{1}{32} (u-1) \sqrt{u} du = \frac{1}{32} \int_{17}^{37} (u^{3/2} - u^{1/2}) du$$

$$= \frac{1}{32} \left(\frac{2}{5} u^{5/2} - \frac{2}{3} u^{3/2} \right) \Big|_{17}^{37} =$$

$$\frac{1}{16} \left(\frac{1}{5} 1369 \sqrt{37} - \frac{1}{3} 37 \sqrt{37} - \frac{1}{5} 289 \sqrt{17} + \frac{1}{3} 17 \sqrt{17} \right)$$

$$= \frac{1}{16} \left(\frac{3822}{15} \sqrt{37} - \frac{782}{15} \sqrt{17} \right)$$

$$= \frac{3822\sqrt{37} - 782\sqrt{17}}{240}$$

$$\begin{array}{r} 122 \\ 1369 \\ \cdot 3 \\ \hline 4107 \\ - 185 \\ \hline 3822 \end{array}$$

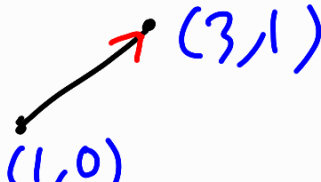
$$\begin{array}{r} 4 \\ 77 \\ \cdot 77 \\ \hline 259 \\ 1110 \\ \hline 1369 \\ \\ 22 \\ 289 \\ - 167 \\ \hline 867 \\ - 85 \\ \hline 782 \end{array}$$

52: Average of $f = \frac{x-1}{1+y^2}$ along C being
line segment from $(1,0)$ to $(3,1)$

Recall that $\text{avg}_C f = \frac{1}{\text{len}(C)} \int_C f ds$, so we need to find $\text{len}(C)$ & $\int_C f ds$.

Step 1: Find length of C & parametrize C

$L = \text{dist}((1,0), (3,1)) = |(3,1) - (1,0)|$
 $= |(2,1)| = \sqrt{2^2 + 1^2} = \sqrt{5}$.



To parametrize, recall $A + t(B-A)$ for $0 \leq t \leq 1$ parametrizes segment from $A \rightarrow B$.

$A = \langle 1, 0 \rangle$, $B = \langle 3, 1 \rangle \Rightarrow \langle 1, 0 \rangle + t \langle 2, 1 \rangle = \langle 1+2t, t \rangle$, $0 \leq t \leq 1$ will be $r(t)$.

Step 2: Find ds , substitute f , eval $\int_C f ds$

$$ds = |r'(t)| dt = |\langle 2, 1 \rangle| dt = \sqrt{5} dt,$$

$$f = \frac{x-1}{1+y^2} = \frac{2t}{1+t^2}, \text{ so } \int_C f ds =$$

$$\int_0^1 \frac{2t}{1+t^2} \sqrt{5} dt = \sqrt{5} \ln(1+t^2) \Big|_0^1 =$$

$$\sqrt{5} \ln 2.$$

Step 3: Combine Step 1 & 2.

$$\text{Average} = \frac{\int_C f ds}{\text{len}(C)} = \frac{\sqrt{5} \ln 2}{\sqrt{5}} = \ln 2.$$

